Long-Term Liabilities

Chapter 14

Chapter 14 Learning Objectives

CONCEPTUAL

C1 Explain the types of notes and prepare entries to account for notes.

C2 *Appendix 14A* – Explain and compute bond pricing.

C3 *Appendix 14C* – Describe accounting for leases and pensions.

ANALYTICAL

A1 Compare bond financing with stock financing.

A2 Assess debt features and their implications.

A3 Compute the debt-to-equity ratio and explain its use.

PROCEDURAL

P1 Prepare entries to record bond issuance and interest expense.

P2 Compute and record amortization of bond discount using straight-line method.

P3 Compute and record amortization of bond premium using straight-line method.

P4 Record the retirement of bonds.

P5 *Appendix 14B* – Compute and record amortization of bond discount using effective interest method.

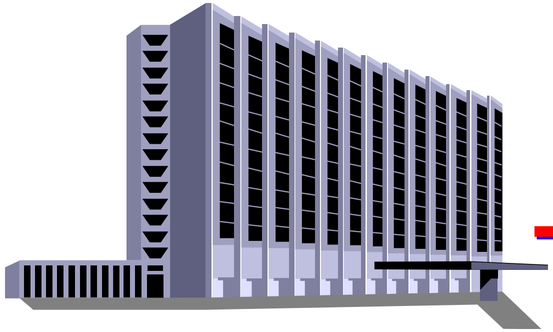
P6 *Appendix 14B* – Compute and record amortization of bond premium using effective interest method.

Learning Objective

A1:
Compare bond financing with stock financing.

Bond Financing

Transactions during the bond life



Corporation

Bond Interest Payments

Investors

Bond Interest Payments

Bond Issue Date

Interest Payment = Bond Par Value × Stated Interest Rate x Time

Learning Objective A1: Compare bond financing with stock financing.

Bond Financing

Advantages

Bonds do not affect owner control.

Interest on bonds is tax deductible.

Bonds can increase return on equity.

Disadvantages

Bonds require payment of both periodic interest and par value at maturity.

Bonds can decrease return on equity.

Learning Objective A1: Compare bond financing with stock financing.

Bond Issuing Procedures

Exhibit 14.2

Learning Objective A1: Compare bond financing with stock financing.

Learning Objective

P1:
Prepare entries to record bond issuance and interest expense.

Issuing Bonds at Par

On Dec. 31, 2017, a company issued the following bonds:

Par Value: $100,000

Stated Interest Rate: 8%

Interest Dates: 6/30 and 12/31

Maturity Date = Dec. 31, 2019 (2 years)

2017			
Dec. 31	Cash	100,000	
	Bonds Payable		100,000
	Sold bonds at par.		

Learning Objective P1: Prepare entries to record bond issuance and interest expense.

Issuing Bonds at Par

On June 30, 2018, the issuer of the bond pays the first semiannual interest payment of $4,000.

2018			
June 30	Bond Interest Expense	4,000	
	Cash		4,000
	Paid semiannual interest (8% × $100,000 × ½ year).		

$100,000 × 8% × ½ year = $4,000

This entry is made every six months until the bonds mature.

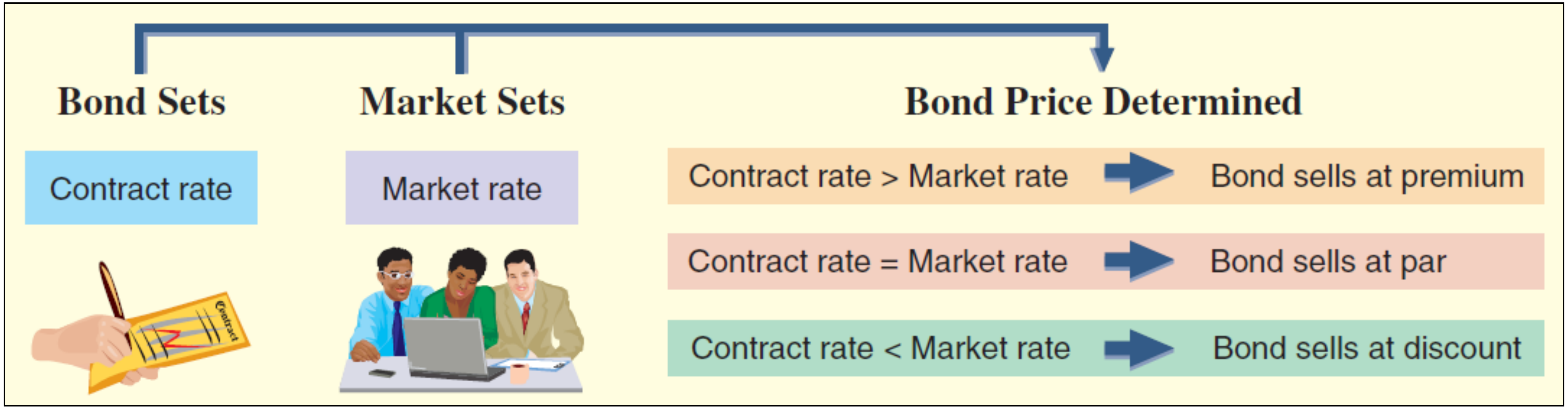
Issuing Bonds at Par

On December 31, 2019, the bonds mature and the issuer of the bond pays face value of $100,000 to the bondholders.

2019			
Dec. 31	Bonds Payable	100,000	
	Cash		100,000
	Paid bond principal at maturity.		

Bond Discount or Premium

Exhibit 14.3



Learning Objective

P2:
Compute and record amortization of bond discount using straight-line method.

Issuing Bonds at a Discount

Fila issues bonds with the following provisions:

Par Value: $100,000

Issue Price: 96.454% of par value

Stated Interest Rate: 8%

Market Interest Rate: 10%

Bond will sell at a discount.

Interest Dates: 6/30 and 12/31

Bond Date: Dec. 31, 2017

Maturity Date: Dec. 31, 2019 (2 years)

Issuing Bonds at a Discount

On Dec. 31, 2017, Fila should record the bond issue.

Par value	$ 100,000
Cash proceeds	96,454 *
Discount	$ 3,546

*$100,000 x 96.454%

Dec. 31	Cash	96,454	
	Discount on Bonds Payable	3,546	
	Bonds Payable		100,000
	Sold bonds at a discount on their issue date.		

Contra-Liability Account

Learning Objective P2: Compute and record amortization of bond discount using straight-line method.

Amortizing a Discount Bond

Straight-Line Amortization Table					
Date	Interest Payment	Interest Expense	Discount Amortization*	Unamortized Discount	Carrying Value
12/31/2017				$ 3,546	$ 96,454
6/30/2018	$ 4,000	$ 4,887	$ 887	2,659	97,341
12/31/2018	4,000	4,887	887	1,772	98,228
6/30/2019	4,000	4,887	887	885	99,115
12/31/2019	4,000	4,885	885	-	100,000
	$ 16,000	$ 19,546	$ 3,546		

* Rounded.

These two columns always sum to par value for a discount bond.

Learning Objective P2: Compute and record amortization of bond discount using straight-line method.

Issuing Bonds at a Discount

Exhibit 14.6

Partial Balance Sheet as of Dec. 31, 2017		
Long-term Liabilities:		
Bonds Payable	100,000	
Less: Discount on Bonds Payable	3,546	96,454

Maturity Value

Carrying Value

Panel A: Interest Computations

Four payments of $4,000 (4 pymts × [$100,000 × 0.08 × 1/2 yr])	$ 16,000
Plus discount	3,546
Total bond interest expense	**$19,546**

$$\text{Bond interest expense (per interest period)} = \frac{\text{Total bond interest expense}}{\text{Number of interest periods}} = \frac{\$19,546}{4} = \$4,887$$

Panel B: Entry to Record Interest Payment and Amortization

2018–2019 June 30 and Dec. 31	Bond Interest Expense	4,887	
	Discount on Bonds Payable		887
	Cash		4,000
	Record semiannual interest and discount amortization (straight-line method).		

Discount ÷ periods

Par value × contract rate

Learning Objective P2: Compute and record amortization of bond discount using straight-line method.

Learning Objective

P3:
Compute and record amortization of bond premium using straight-line method.

Issuing Bonds at a Premium

Adidas issues bonds with the following provisions:

Par Value: $100,000

Issue Price: 103.546% of par value

Stated Interest Rate: 12%

Market Interest Rate: 10%

Bond will sell at a premium.

Interest Dates: 6/30 and 12/31

Bond Date: Dec. 31, 2017

Maturity Date: Dec. 31, 2019 (2 years)

Learning Objective P3: Compute and record amortization of bond premium using straight-line method.

Issuing Bonds at a Premium

On Dec. 31, 2017, Adidas will record the bond issue as:

Par value	$ 100,000
Cash proceeds	103,546*
Premium	$ 3,546

*$100,000 x 103.546%

Dec. 31	Cash	103,546	
	Premium on Bonds Payable		3,546
	Bonds Payable.		100,000
	Sold bonds at a premium on their issue date.		

Adjunct-Liability Account

Learning Objective P3: Compute and record amortization of bond premium using straight-line method.

Issuing Bonds at a Premium

Partial Balance Sheet as of Dec. 31, 2017		
Long-term Liabilities:		
Bonds Payable	100,000	
Plus: Premum on Bonds Payable	3,546	103,546

Maturity Value

Carrying Value

Amortizing a Premium Bond

Adidas will make the following entry every six months to record the cash interest payment and the amortization of the discount.

June 30 and Dec. 31	Bond Interest Expense	5,113	
	Premium on Bonds Payable	887	
	Cash		6,000
	To record semiannual interest and premium amortization (straight-line method).		

\$3,546 ÷ 4 periods = \$887 (rounded)

\$100,000 × 12% × ½ = \$6,000

Learning Objective P3: Compute and record amortization of bond premium using straight-line method.

Amortizing a Premium Bond

Straight-Line Amortization Table					
Date	Interest Payment	Interest Expense	Premium Amortization*	Unamortized Premium	Carrying Value
12/31/2017				$ 3,546	$ 103,546
6/30/2018	$ 6,000	$ 5,113	$ 887	2,659	102,659
12/31/2018	6,000	5,113	887	1,772	101,772
6/30/2019	6,000	5,113	887	885	100,885
12/31/2019	6,000	5,115	885	-	100,000
	$ 24,000	$ 20,454	$ 3,546		

* Rounded.

Learning Objective P3: Compute and record amortization of bond premium using straight-line method.

Learning Objective

P4:
Record the retirement of bonds.

Bond Retirement

Retirement of the Fila bonds at maturity for $100,000 cash.

Dec. 31	Bonds Payable	100,000	
	Cash		100,000
	To record retirement of bonds at maturity.		

Because any discount or premium will be fully amortized at maturity, the carrying value of the bonds will be equal to par value.

Learning Objective P4: Record the retirement of bonds.

Bond Retirement before Maturity

Carrying Value > Retirement Price = Gain

Carrying Value < Retirement Price = Loss

Assume that $100,000 of callable bonds will be retired on July 1, 2017, after the first interest payment. The bond carrying value is $104,500.The bonds have a call premium of $3,000.

July 1	Bonds Payable	100,000	
	Premium on Bonds Payable	4,500	
	Gain on Bond Retirement		1,500
	Cash		103,000
	To record retirement of bonds before maturity.		

Learning Objective P4: Record the retirement of bonds.

Bond Retirement

Conversion of Bonds to Stock

On January 1, $100,000 par value bonds of Converse, with a carrying value of $100,000, are converted to 15,000 shares of $2 par value common stock.

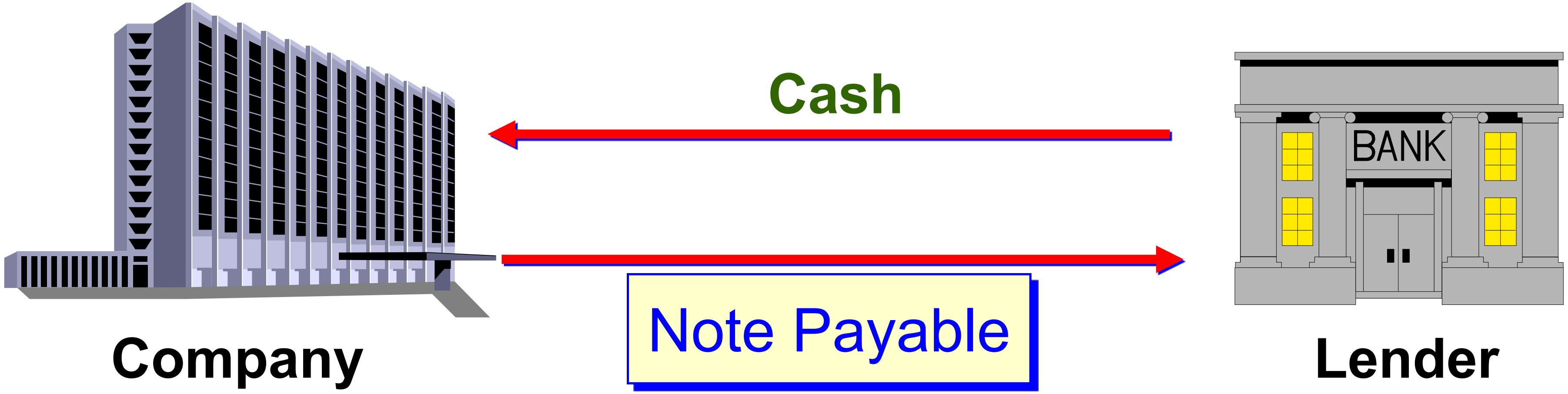
Jan. 1	Bonds Payable	100,000	
	Common Stock		30,000
	Paid-In Capital in Excess of Par Value		70,000
	To record retirement of bonds by conversion.		

15,000 shares × $2 par value per share

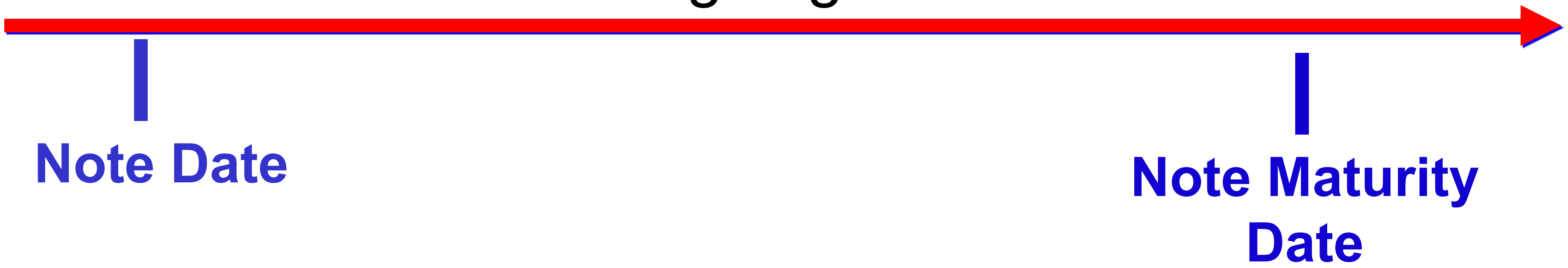
Learning Objective

C1:
Explain the types of notes and prepare entries to account for notes.

Long-Term Notes Payable

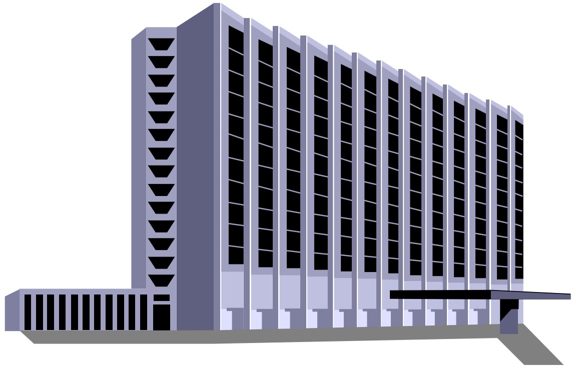


When is the repayment of the principal and interest going to be made?

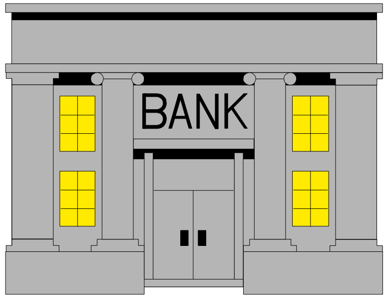


Learning Objective C1: Explain the types of notes and prepare entries to account for notes.

Long-Term Notes Payable



Single Payment of Principal plus Interest



Company

Lender

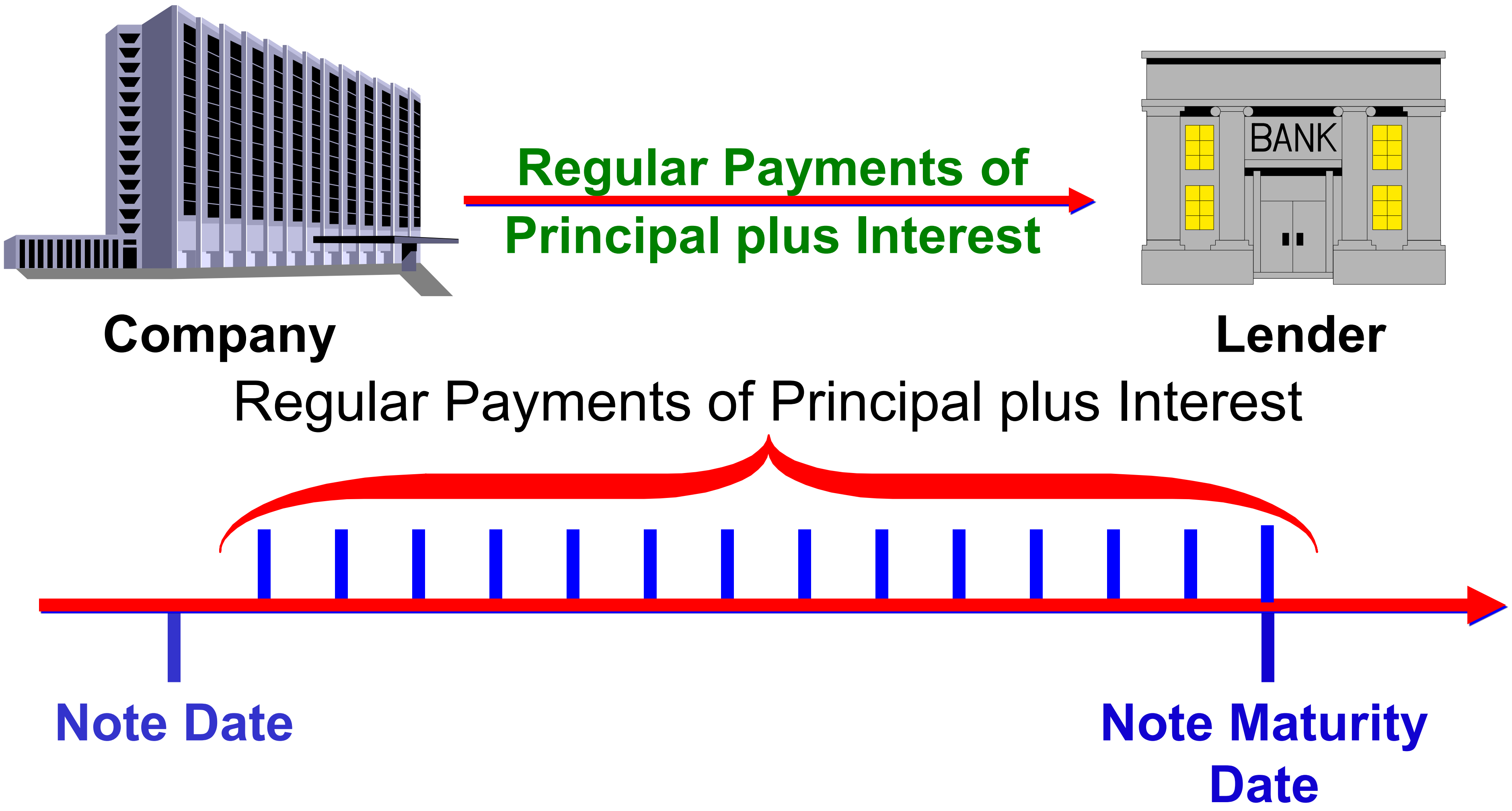
Single Payment of Principal plus Interest

Note Date

Note Maturity Date

Learning Objective C1: Explain the types of notes and prepare entries to account for notes.

Long-Term Notes Payable



Learning Objective C1: Explain the types of notes and prepare entries to account for notes.

Installment Notes

On January 1, 2017 Foghog borrows $60,000 from a bank to purchase equipment. It signs an 8% installment note requiring 3 annual payments of principal plus interest.

Jan. 1	Cash	60,000	
	Notes Payable		60,000
	Borrowed $60,000 by signing 8%, three-year note.		

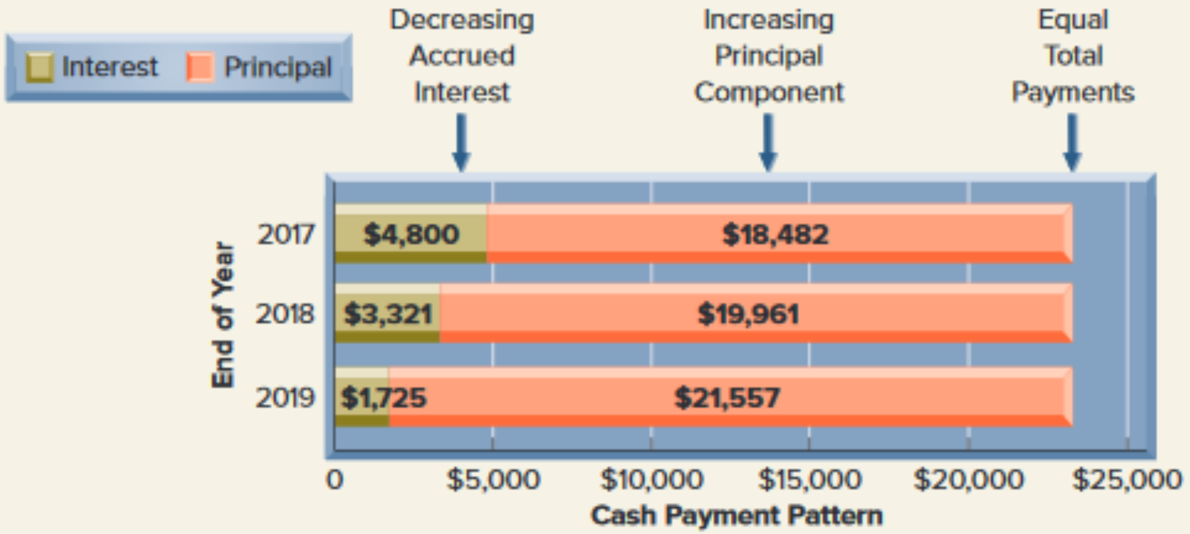
Compute the periodic payment by dividing the face amount of the note by the present value factor.

Computation	Table	Table Value	Present Value	Payment
Principal divided by PV factor	Annuity of $1 (B.3)	2.5771	60,000	23,282

Installment Notes with Equal Payments

Exhibit 14.12

			Payments				
Period Ending Date	(A) Beginning Balance	(B) *Debit* Interest Expense 8% × (A)	+	(C) *Debit* Notes Payable (D) – (B)	=	(D) *Credit* Cash (computed)	(E) Ending Balance (A) – (C)
(1) **12/31/2017**	**$60,000**	**$4,800**		**$ 18,482**		**$23,282**	**$41,518**
(2) 12/31/2018	41,518	3,321		19,961		23,282	21,557
(3) 12/31/2019	21,557	1,725		21,557		23,282	0
		$9,846		**$60,000**		**$69,846**	



Learning Objective C1: Explain the types of notes and prepare entries to account for notes.

Installment Notes with Equal Payments

Let's record the first payment made on December 31, 2017 by Foghog to the bank.

2017			
Dec. 31	Interest Expense	4,800	
	Notes Payable	18,482	
	Cash		23,282
	Record first installment payment.		

Refer back to the amortization schedule to make the December 31, 2018 payment on the note.

2018			
Dec. 31	Interest Expense	3,321	
	Notes Payable	19,961	
	Cash		23,282
	Record second installment payment.		

Mortgage Notes and Bonds

A mortgage is a legal agreement that helps protect the lender if the borrower fails to make the required payments.

It gives the lender the right to be paid out of the cash proceeds from the sale of the borrower's assets specifically identified in the mortgage contract.

Learning Objective

A2:
Compare bond financing with stock financing.

Features of Bonds and Notes

Secured and Unsecured

Convertible and Callable

Term and Serial

Registered and Bearer

Learning Objective A2: Compare bond financing with stock financing.

End of Chapter 14

Learning Objective

A3:
Compute the
debt-to-equity ratio and
explain its use.

Debt-to-Equity Ratio

Exhibit 14.14

$$\text{Debt-to-equity ratio} = \frac{\text{Total liabilities}}{\text{Total equity}}$$

This ratio helps investors determine the risk of investing in a company by dividing its total liabilities by total equity.

$ millions	2015	2014	2013	2012
Total liabilities	$52,060	$43,764	$30,413	$24,363
Total equity	$13,384	$10,741	$ 9,746	$ 8,192
Debt-to-equity	3.9	4.1	3.1	3.0
Industry debt-to-equity	1.7	1.8	1.6	1.5

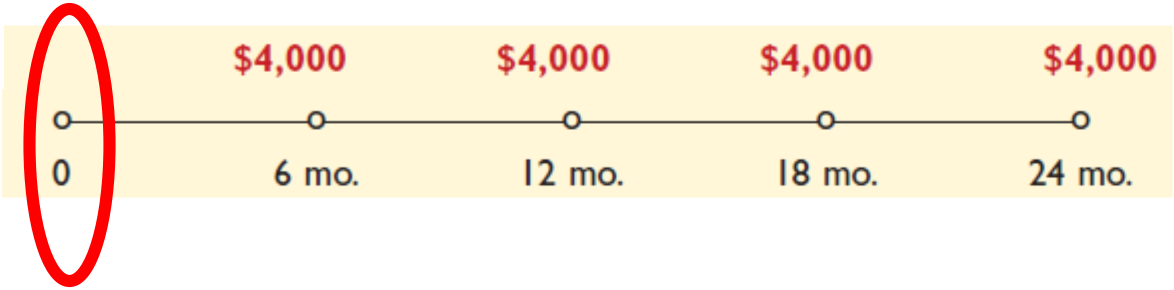
Learning Objective A3: Compute the debt-to-equity ratio and explain its use.

Learning Objective

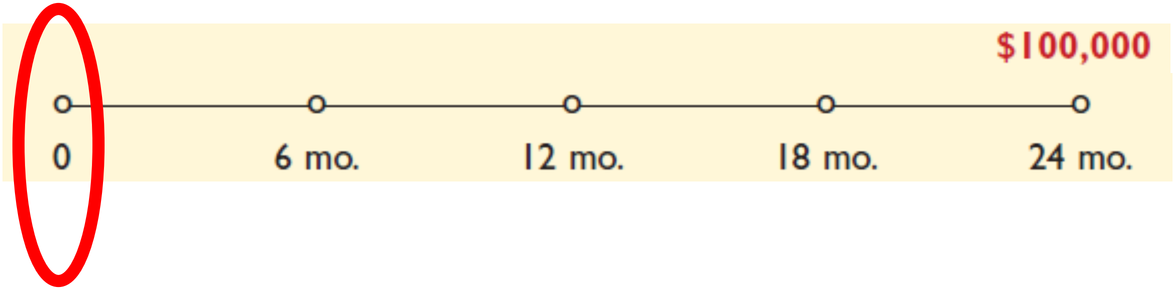
C2: *Appendix 14A*
Explain and compute bond pricing.

Bond Pricing

Cash Outflows related to Interest Payments



Cash Outflows for par value at end of Bond life



Learning Objective C2: Explain and compute bond pricing.

Present Value of Discount Bond

Fila issues bonds with the following provisions:

Par Value: $100,000

Issue Price: ?

Stated Interest Rate: 8%

Market Interest Rate: 10%

Interest Dates: 6/30 and 12/31

Bond Date: Dec. 31, 2017

Maturity Date: Dec. 31, 2019 (2 years)

Learning Objective C2: Explain and compute bond pricing.

Present Value of Discount Bond

To calculate Present Value, we need relevant interest rate and number of periods.
Semiannual rate = 5% (Market rate 10% ÷ 2)
Semiannual periods = 4 (Bond life 2 years × 2)

Exhibit 14A.1

Cash Flow	Table	Present Value Factor		Amount		Present Value
$100,000 par (maturity) value	B.1	0.8227	×	$100,000	=	$ 82,270
$4,000 interest payments	B.3	3.5460	×	4,000	=	14,184
Price of bond						$96,454

$100,000 × 8% × ½ = $4,000

Learning Objective C2: Explain and compute bond pricing.

Present Value of Premium Bond

To calculate Present Value, we need relevant interest rate and number of periods.
Semiannual rate = 5% (Market rate 10% ÷ 2)
Semiannual periods = 4 (Bond life 2 years × 2)

Exhibit 14A.2

Cash Flow	Table	Present Value Factor		Amount		Present Value
$100,000 par (maturity) value	B.1 (PV of 1)	0.8227	×	$100,000	=	$ 82,270
$6,000 interest payments..............	B.3 (PV of ann.)	3.5460	×	6,000	=	21,276
Price of bond......................						$103,546

$100,000 × 12% × ½ = $6,000

Learning Objective C2: Explain and compute bond pricing.

Learning Objective

P5: *Appendix 14B*
Compute and record amortization of bond discount using effective interest method.

Appendix 14B: Effective Interest Amortization

Exhibit 14B.1

	A	B	C	D	E	F
1–2	Bonds: $100,000 Par Value, Semiannual Interest Payments, Two-Year Life, 4% Semiannual Contract Rate, 5% Semiannual Market Rate					
3–6	Semiannual Interest Period-End	(A) Cash Interest Paid 4% × $100,000	(B) Bond Interest Expense 5% × Prior (E)	(C) Discount Amortization (B) – (A)	(D) Unamortized Discount Prior (D) – (C)	(E) Carrying Value $100,000 – (D)
7	(0) 12/31/2017				$3,546	$ 96,454
8	(1) 6/30/2018	$ 4,000	$ 4,823	$ 823	2,723	97,277
9	(2) 12/31/2018	4,000	4,864	864	1,859	98,141
10	(3) 6/30/2019	4,000	4,907	907	952	99,048
11	(4) 12/31/2019	4,000	4,952	952	0	100,000
12		$16,000	$19,546	$3,546		

Column (A) is the par value ($100,000) multiplied by the semiannual contract rate (4%).
Column (B) is the prior period's carrying value multiplied by the semiannual market rate (5%).
Column (C) is the difference between interest paid and bond interest expense, or [(B) – (A)].
Column (D) is the prior period's unamortized discount less the current period's discount amortization.
Column (E) is the par value less unamortized discount, or [$100,000 – (D)].

Except for differences in amounts, journal entries recording the expense and updating the liability balance are the same under the effective interest method and the straight-line method. We can use the numbers in Exhibit 14B.1 to record each semiannual entry during the bonds' two-year life (June 30, 2018, through December 31, 2019). For instance, we record the interest payment at the end of the first semiannual period as follows.

2018			
June 30	Bond Interest Expense	4,823	
	Discount on Bonds Payable		823
	Cash		4,000
	Record semiannual interest and discount amortization (effective interest method).		

Effective Interest Amortization of Bond Discount

Stated Rate: 8%

Effective Rate: 10%

Learning Objective P5: Compute and record amortization of bond discount using effective interest method.

Learning Objective

P6: *Appendix 14B*
Compute and record amortization of bond premium using effective interest method.

Appendix 14B: Effective Interest Amortization

Exhibit 14B.2

	A		B	C	D	E	F
1–2	Bonds: $100,000 Par Value, Semiannual Interest Payments, Two-Year Life, 6% Semiannual Contract Rate, 5% Semiannual Market Rate						
3–6	Semiannual Interest Period-End		(A) Cash Interest Paid 6% × $100,000	(B) Bond Interest Expense 5% × Prior (E)	(C) Premium Amortization (A) – (B)	(D) Unamortized Premium Prior (D) – (C)	(E) Carrying Value $100,000 + (D)
7	(0)	12/31/2017				$3,546	$103,546
8	(1)	6/30/2018	$ 6,000	$ 5,177	$ 823	2,723	102,723
9	(2)	12/31/2018	6,000	5,136	864	1,859	101,859
10	(3)	6/30/2019	6,000	5,093	907	952	100,952
11	(4)	12/31/2019	6,000	5,048	952	0	100,000
12			$24,000	$20,454	$3,546		

Column (A) is the par value ($100,000) multiplied by the semiannual contract rate (6%).
Column (B) is the prior period's carrying value multiplied by the semiannual market rate (5%).
Column (C) is the difference between interest paid and bond interest expense, or [(A) – (B)].
Column (D) is the prior period's unamortized premium less the current period's premium amortization.
Column (E) is the par value plus unamortized premium, or [$100,000 + (D)].

When the issuer makes the first semiannual interest payment, the effect of premium amortization on bond interest expense and bond liability is recorded as follows.

2018			
June 30	Bond Interest Expense	5,177	
	Premium on Bonds Payable	823	
	Cash		6,000
	Record semiannual interest and premium amortization (effective interest method).		

Effective Interest Amortization of Bond Premium

Stated Rate: 12%

Effective Rate: 10%

Learning Objective P6: Compute and record amortization of bond premium using effective interest method.

Learning Objective

C3: *Appendix 14C*

Describe accounting for leases and pensions.

Appendix 14C: Leases and Pensions

A lease is a contractual agreement between the lessor (asset owner) and the lessee (asset renter or tenant) that grants the lessee the right to use the asset for a period of time in return for cash (rent) payments.

Operating Leases

Operating leases are short-term (or cancelable) leases in which the lessor retains the risks and rewards of ownership. Examples include most car and apartment rental agreements.

Capital Leases

Capital leases are long-term (or non-cancelable) leases by which the lessor transfers substantially all risks and rewards of ownership to the lessee. Examples include leases of airplanes and department store buildings.

Learning Objective C3: Describe accounting for leases and pensions.

Appendix 14C: Leases and Pensions

A pension is a contractual agreement between an employer and its employees for the employer to provide benefits (payments) to employees after they retire.

Defined Benefit Plans

The employer's contributions vary, depending on assumptions about future pension assets and liabilities. A pension liability is reported when the accumulated benefit obligation is more than the plan assets, a so-called underfunded plan.

NEED-TO-KNOW 14-1

A company issues 8%, two-year bonds on December 31, 20X1, with a par value of $7,000 and semiannual interest payments. On the issue date, the annual market rate for these bonds is 8%, which implies a selling price of $7,000. Prepare journal entries to record (a) the issuance of bonds on December 31, 20X1; (b) the first through fourth interest payments on each June 30 and December 31; and (c) the maturity of the bond on December 31, 20X3.

	General Journal		Debit	Credit
Issuance	Cash		7,000	
	Bonds payable			7,000
Every 6 mos.	Bond interest expense	($7,000 x .08 market / 2)	280	
	Cash	($7,000 x .08 contract / 2)		280
Maturity	Bonds payable		7,000	
	Cash			7,000

Learning Objective P1: Prepare entries to record bond issuance and interest expense.

NEED-TO-KNOW 14-2

A company issues 8%, two-year bonds on December 31, 20X1, with a par value of $7,000 and semiannual interest payments. On the issue date, the annual market rate for these bonds is 10%, which implies a selling price of 96.46 or $6,752.

(a) Prepare an amortization table for these bonds; use the straight-line method to amortize the discount. Then, prepare journal entries to record (b) the issuance of bonds on December 31, 20X1; (c) the first through fourth interest payments on each June 30 and December 31; and (d) the maturity of the bond on December 31, 20X3.

Change in Carrying Value
$248 / 4 semiannual periods = $62 per period

Semiannual Period-End		Unamortized Discount	Carrying Value	
(0)	12/31/20X1	$248	$6,752	<< $7,000 x .9646 = $6,752
(1)	06/30/20X2	186	6,814	
(2)	12/31/20X2	124	6,876	
(3)	06/30/20X3	62	6,938	
(4)	12/31/20X3	0	$7,000	

Discount on Bonds Payable

12/31/20X1	248		
		06/30/20X2	62
		12/31/20X2	62
		06/30/20X3	62
		12/31/20X3	62
12/31/20X3	0		

Bonds Payable

		12/31/20X1	7,000
		12/31/20X3	7,000

Learning Objective P1: Prepare entries to record bond issuance and interest expense.

Learning Objective P2: Compute and record amortization of bond discount using straight-line method.

NEED-TO-KNOW 14-2

A company issues 8%, two-year bonds on December 31, 20X1, with a par value of $7,000 and semiannual interest payments. On the issue date, the annual market rate for these bonds is 10%, which implies a selling price of 96.46 or $6,752.

Discount on Bonds Payable

12/31/20X1	248		
		06/30/20X2	62
		12/31/20X2	62
		06/30/20X3	62
		12/31/20X3	62
12/31/20X3	0		

Bonds Payable

		12/31/20X1	7,000
		12/31/20X3	7,000

	General Journal	Debit	Credit
12/31/20X1	Cash ($7,000 x .9646)	6,752	
	Discount on Bonds Payable	248	
	Bonds Payable		7,000

Learning Objective P1: Prepare entries to record bond issuance and interest expense.

Learning Objective P2: Compute and record amortization of bond discount using straight-line method.

NEED-TO-KNOW 14-2

A company issues 8%, two-year bonds on December 31, 20X1, with a par value of $7,000 and semiannual interest payments. On the issue date, the annual market rate for these bonds is 10%, which implies a selling price of 96.46 or $6,752.

Discount on Bonds Payable

Date	Debit	Date	Credit
12/31/20X1	248		
		06/30/20X2	62
		12/31/20X2	62
		06/30/20X3	62
		12/31/20X3	62
12/31/20X3	0		

Bonds Payable

Date	Debit	Date	Credit
		12/31/20X1	7,000
		12/31/20X3	7,000

	General Journal	Debit	Credit
06/30/20X2	Bond Interest Expense ($280 + $62)	342	
	Discount on Bonds Payable		62
	Cash ($7,000 x 8% / 2)		280
12/31/20X2	Bond Interest Expense ($280 + $62)	342	
	Discount on Bonds Payable		62
	Cash ($7,000 x 8% / 2)		280
06/30/20X3	Bond Interest Expense ($280 + $62)	342	
	Discount on Bonds Payable		62
	Cash ($7,000 x 8% / 2)		280
12/31/20X3	Bond Interest Expense ($280 + $62)	342	
	Discount on Bonds Payable		62
	Cash ($7,000 x 8% / 2)		280

Learning Objective P1: Prepare entries to record bond issuance and interest expense.

Learning Objective P2: Compute and record amortization of bond discount using straight-line method.

NEED-TO-KNOW 14-2

A company issues 8%, two-year bonds on December 31, 20X1, with a par value of $7,000 and semiannual interest payments. On the issue date, the annual market rate for these bonds is 10%, which implies a selling price of 96.46 or $6,752.

Interest expense = Amount repaid minus amount borrowed

4 payments of $280	$1,120	
1 payment of $7,000	7,000	
Total repaid	8,120	
Borrowed	6,752	
Total bond interest expense	$1,368	÷4 = $342

	General Journal	Debit	Credit
06/30/20X2	Bond Interest Expense ($280 + $62)	342	
	Discount on Bonds Payable		62
	Cash ($7,000 x 8% / 2)		280
12/31/20X2	Bond Interest Expense ($280 + $62)	342	
	Discount on Bonds Payable		62
	Cash ($7,000 x 8% / 2)		280
06/30/20X3	Bond Interest Expense ($280 + $62)	342	
	Discount on Bonds Payable		62
	Cash ($7,000 x 8% / 2)		280
12/31/20X3	Bond Interest Expense ($280 + $62)	342	
	Discount on Bonds Payable		62
	Cash ($7,000 x 8% / 2)		280

Learning Objective P1: Prepare entries to record bond issuance and interest expense.

Learning Objective P2: Compute and record amortization of bond discount using straight-line method.

NEED-TO-KNOW 14-2

A company issues 8%, two-year bonds on December 31, 20X1, with a par value of $7,000 and semiannual interest payments. On the issue date, the annual market rate for these bonds is 10%, which implies a selling price of 96.46 or $6,752.

Discount on Bonds Payable

12/31/20X1	248		
		06/30/20X2	62
		12/31/20X2	62
		06/30/20X3	62
		12/31/20X3	62
12/31/20X3	0		

Bonds Payable

		12/31/20X1	7,000
		12/31/20X3	7,000

	General Journal	Debit	Credit
12/31/20X3	Bonds Payable	7,000	
	Cash		7,000

Learning Objective P1: Prepare entries to record bond issuance and interest expense.

Learning Objective P2: Compute and record amortization of bond discount using straight-line method.

NEED-TO-KNOW 14-3

A company issues 8%, two-year bonds on December 31, 20X1, with a par value of $7,000 and semiannual interest payments. On the issue date, the annual market rate for these bonds is 6%, which implies a selling price of 103.71 or $7,260.

(a) Prepare an amortization table for these bonds; use the straight-line method to amortize the premium. Then, prepare journal entries to record (b) the issuance of bonds on December 31, 20X1; (c) the first through fourth interest payments on each June 30 and December 31; and (d) the maturity of the bond on December 31, 20X3.

Change in Carrying Value
$260 / 4 semiannual periods = $65 per period

Semiannual Period-End		Unamortized Premium	Carrying Value	
(0)	12/31/20X1	$260	$7,260	<< $7,000 x 1.0371 = $7,260
(1)	06/30/20X2	195	7,195	
(2)	12/31/20X2	130	7,130	
(3)	06/30/20X3	65	7,065	
(4)	12/31/20X3	0	$7,000	

Bonds Payable

Debit		Credit	
		12/31/20X1	7,000
		12/31/20X3	7,000

Premium on Bonds Payable

Debit		Credit	
		12/31/20X1	260
06/30/20X2	65		
12/31/20X2	65		
06/30/20X3	65		
12/31/20X3	65		
		12/31/20X3	0

Learning Objective P3: Compute and record amortization of bond premium using straight-line method.

NEED-TO-KNOW 14-3

A company issues 8%, two-year bonds on December 31, 20X1, with a par value of $7,000 and semiannual interest payments. On the issue date, the annual market rate for these bonds is 6%, which implies a selling price of 103.71 or $7,260. Semiannual payment = $280 ($7,000 x 8% x ½)

Bonds Payable

Debit		Credit	
		12/31/20X1	7,000
		12/31/20X3	7,000

Premium on Bonds Payable

Debit		Credit	
		12/31/20X1	260
06/30/20X2	65		
12/31/20X2	65		
06/30/20X3	65		
12/31/20X3	65		
		12/31/20X3	0

	General Journal	Debit	Credit
12/31/20X1	Cash ($7,000 x 1.0371)	7,260	
	Premium on Bonds Payable		260
	Bonds Payable		7,000

Interest expense = Amount repaid minus amount borrowed

4 payments of $280	$1,120	
1 payment of $7,000	7,000	
Total repaid	8,120	
Borrowed	7,260	
Total bond interest expense	$ 860	÷4 = $215 per period

Learning Objective P3: Compute and record amortization of bond premium using straight-line method.

NEED-TO-KNOW 14-3

A company issues 8%, two-year bonds on December 31, 20X1, with a par value of $7,000 and semiannual interest payments. On the issue date, the annual market rate for these bonds is 6%, which implies a selling price of 103.71 or $7,260. Semiannual payment = $280 ($7,000 x 8% x ½)

Bonds Payable

		12/31/20X1	7,000
		12/31/20X3	7,000

Premium on Bonds Payable

		12/31/20X1	260
06/30/20X2	65		
12/31/20X2	65		
06/30/20X3	65		
12/31/20X3	65		
		12/31/20X3	0

	General Journal	Debit	Credit
06/30/20X2	Bond Interest Expense ($280 - $65)	215	
	Premium on Bonds Payable	65	
	Cash ($7,000 x 8% x ½)		280
12/31/20X2	Bond Interest Expense ($280 - $65)	215	
	Premium on Bonds Payable	65	
	Cash ($7,000 x 8% x ½)		280
06/30/20X3	Bond Interest Expense ($280 - $65)	215	
	Premium on Bonds Payable	65	
	Cash ($7,000 x 8% x ½)		280
12/31/20X3	Bond Interest Expense ($280 - $65)	215	
	Premium on Bonds Payable	65	
	Cash ($7,000 x 8% x ½)		280

Learning Objective P3: Compute and record amortization of bond premium using straight-line method.

NEED-TO-KNOW 14-3

A company issues 8%, two-year bonds on December 31, 20X1, with a par value of $7,000 and semiannual interest payments. On the issue date, the annual market rate for these bonds is 6%, which implies a selling price of 103.71 or $7,260.   Semiannual payment = $280 ($7,000 x 8% x ½)

Bonds Payable

		12/31/20X1	7,000
		12/31/20X3	7,000

Premium on Bonds Payable

		12/31/20X1	260
06/30/20X2	65		
12/31/20X2	65		
06/30/20X3	65		
12/31/20X3	65		
		12/31/20X3	0

	General Journal	Debit	Credit
12/31/20X3	Bonds Payable	7,000	
	Cash		7,000

Learning Objective P3: Compute and record amortization of bond premium using straight-line method.

NEED-TO-KNOW 14-4

On January 1, 20X1, a company borrows $1,000 cash by signing a four-year, 5% installment note. The note requires four equal total payments of accrued interest and principal on December 31 of each year from 20X1 through 20X4.

1. Compute the amount of each of the four equal total payments.

$1,000 = Present Value of an Ordinary Annuity

$1,000 = PVA (n=4, i=5%) x Payment

$$\frac{\$1{,}000}{\text{PVA }(n=4,\ i=5\%)} = \text{Payment}$$

PV of $1

FV of $1

PV Ord Ann

FV Ord Ann

Learning Objective P5: Record the retirement of bonds.

Learning Objective C1: Explain the types of notes and prepare entries to account for notes.

NEED-TO-KNOW 14-4

$$p = \left[1 - \frac{1}{(1+i)^n}\right] / i$$

TABLE B.3

Present Value of an Annuity of 1

Periods	Rate 1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	12%	15%
1	0.9901	0.9804	0.9709	0.9615	0.9524	0.9434	0.9346	0.9259	0.9174	0.9091	0.8929	0.8696
2	1.9704	1.9416	1.9135	1.8861	1.8594	1.8334	1.8080	1.7833	1.7591	1.7355	1.6901	1.6257
3	2.9410	2.8839	2.8286	2.7751	2.7232	2.6730	2.6243	2.5771	2.5313	2.4869	2.4018	2.2832
4	3.9020	3.8077	3.7171	3.6299	3.5460	3.4651	3.3872	3.3121	3.2397	3.1699	3.0373	2.8550
5	4.8534	4.7135	4.5797	4.4518	4.3295	4.2124	4.1002	3.9927	3.8897	3.7908	3.6048	3.3522
6	5.7955	5.6014	5.4172	5.2421	5.0757	4.9173	4.7665	4.6229	4.4859	4.3553	4.1114	3.7845
7	6.7282	6.4720	6.2303	6.0021	5.7864	5.5824	5.3893	5.2064	5.0330	4.8684	4.5638	4.1604
8	7.6517	7.3255	7.0197	6.7327	6.4632	6.2098	5.9713	5.7466	5.5348	5.3349	4.9676	4.4873
9	8.5660	8.1622	7.7861	7.4353	7.1078	6.8017	6.5152	6.2469	5.9952	5.7590	5.3282	4.7716
10	9.4713	8.9826	8.5302	8.1109	7.7217	7.3601	7.0236	6.7101	6.4177	6.1446	5.6502	5.0188
11	10.3676	9.7868	9.2526	8.7605	8.3064	7.8869	7.4987	7.1390	6.8052	6.4951	5.9377	5.2337
12	11.2551	10.5753	9.9540	9.3851	8.8633	8.3838	7.9427	7.5361	7.1607	6.8137	6.1944	5.4206
13	12.1337	11.3484	10.6350	9.9856	9.3936	8.8527	8.3577	7.9038	7.4869	7.1034	6.4235	5.5831
14	13.0037	12.1062	11.2961	10.5631	9.8986	9.2950	8.7455	8.2442	7.7862	7.3667	6.6282	5.7245
15	13.8651	12.8493	11.9379	11.1184	10.3797	9.7122	9.1079	8.5595	8.0607	7.6061	6.8109	5.8474
16	14.7179	13.5777	12.5611	11.6523	10.8378	10.1059	9.4466	8.8514	8.3126	7.8237	6.9740	5.9542
17	15.5623	14.2919	13.1661	12.1657	11.2741	10.4773	9.7632	9.1216	8.5436	8.0216	7.1196	6.0472
18	16.3983	14.9920	13.7535	12.6593	11.6896	10.8276	10.0591	9.3719	8.7556	8.2014	7.2497	6.1280
19	17.2260	15.6785	14.3238	13.1339	12.0853	11.1581	10.3356	9.6036	8.9501	8.3649	7.3658	6.1982
20	18.0456	16.3514	14.8775	13.5903	12.4622	11.4699	10.5940	9.8181	9.1285	8.5136	7.4694	6.2593
25	22.0232	19.5235	17.4131	15.6221	14.0939	12.7834	11.6536	10.6748	9.8226	9.0770	7.8431	6.4641
30	25.8077	22.3965	19.6004	17.2920	15.3725	13.7648	12.4090	11.2578	10.2737	9.4269	8.0552	6.5660
35	29.4086	24.9986	21.4872	18.6646	16.3742	14.4982	12.9477	11.6546	10.5668	9.6442	8.1755	6.6166
40	32.8347	27.3555	23.1148	19.7928	17.1591	15.0463	13.3317	11.9246	10.7574	9.7791	8.2438	6.6418

Learning Objective P5: Record the retirement of bonds.

Learning Objective C1: Explain the types of notes and prepare entries to account for notes.

NEED-TO-KNOW 14-4

On January 1, 20X1, a company borrows $1,000 cash by signing a four-year, 5% installment note. The note requires four equal total payments of accrued interest and principal on December 31 of each year from 20X1 through 20X2

1. Compute the amount of each of the four equal total payments.

$1,000 = Present Value of an Ordinary Annuity

$1,000 = PVA (n=4, i=5%) x Payment

$$\frac{\$1{,}000}{\text{PVA }(n=4,\ i=5\%)} = \text{Payment}$$

$$\frac{\$1{,}000}{3.5460} = \text{Payment}$$

$282 = Payment

PV of $1

FV of $1

PV Ord Ann

FV Ord Ann

Learning Objective P5: Record the retirement of bonds.

Learning Objective C1: Explain the types of notes and prepare entries to account for notes.

NEED-TO-KNOW 14-4

On January 1, 20X1, a company borrows $1,000 cash by signing a four-year, 5% installment note. The note requires four equal total payments of accrued interest and principal on December 31 of each year from 20X1 through 20X4.

1. Compute the amount of each of the four equal total payments. $282

2. Prepare an amortization table for this installment note.

Period End	(A) Beginning Balance	(B) Debit Interest Expense [5% x (A)]	(C) Debit Notes Payable [(D) - (B)]	(D) Credit Cash	(E) Ending Balance [(A) - (C)]
12/31/20X1	$1,000	$50	$232	$282	$768
12/31/20X2	768	38	244	282	524
12/31/20X3	524	26	256	282	268
12/31/20X4	268	14	268	282	0
		$128	$1,000	$1,128	

3. Prepare journal entries to record the loan on January 1, 20X1, and the four payments from December 31, 20X1, through December 31, 20X4.

Learning Objective P5: Record the retirement of bonds.

Learning Objective C1: Explain the types of notes and prepare entries to account for notes.

NEED-TO-KNOW 14-4

Period End	(A) Beginning Balance	(B) Debit Interest Expense [5% x (A)]	(C) Debit Notes Payable [(D) - (B)]	(D) Credit Cash	(E) Ending Balance [(A) - (C)]
12/31/20X1	$1,000	$50	$232	$282	$768
12/31/20X2	768	38	244	282	524
12/31/20X3	524	26	256	282	268
12/31/20X4	268	14*	268	282	0

	General Journal		Debit	Credit
01/01/20X1	Cash		1,000	
	Notes payable			1,000
12/31/20X1	Interest expense	($1,000 x .05)	50	
	Notes payable		232	
	Cash			282
12/31/20X2	Interest expense	($768 x .05)	38	
	Notes payable		244	
	Cash			282
12/31/20X3	Interest expense	($524 x .05)	26	
	Notes payable		256	
	Cash			282
12/31/20X4	Interest expense	(* Rounded)	14	
	Notes payable		268	
	Cash			282

Learning Objective P5: Record the retirement of bonds.

Learning Objective C1: Explain the types of notes and prepare entries to account for notes.